

Codebook: floodopioid_demo.dta

County-year panel of 420 Appalachian counties, 1999-2017 (N = 7,980).

Identifiers

Variable	Description
year	Calendar year (1999-2017)
yearstr	Year as string
state	State abbreviation (2-letter)
state_fips	State FIPS code
county_fips	County FIPS code (5-digit)

Population Variables

Source: U.S. Census Bureau Population Estimates Program

Variable	Description
pop_allpop	Total population
pop_male	Male population
pop_female	Female population
pop_white	White population
pop_nonwhite	Non-white population
pop_less25	Population under 25
pop_2544	Population 25-44
pop_4564	Population 45-64
pop_over65	Population 65+
pop_leh	Low-education population (less than high school)
pop_hh	High-education population (bachelor's or higher)

Mortality Outcomes

Source: CDC WONDER (restricted-use mortality files)

All death variables are annual county-level counts.

Variable	Description
death_opioid_{pop}	Opioid overdose death count for subgroup {pop}
death_alldrug_{pop}	All drug overdose death count for subgroup {pop}
death_pre_{pop}	Prescription drug overdose death count for subgroup {pop}

Where {pop} is one of: allpop, male, female, white, nonwhite, leh, hh, less25, 2544, 4564, over65

Socioeconomic Controls

Source: American Community Survey 5-Year Estimates, BEA

Variable	Description
unemployrate	Unemployment rate (%)
povertypct	Poverty rate (%)
medianhhi	Median household income (\$)
edupct_leh	% population with less than high school education (available 2008+)
edupct_hh	% population with bachelor's degree or higher (available 2008+)

Migration

Source: U.S. Census Bureau Population Estimates Program

Variable	Description
rnetmig	Net migration rate (per 1,000 population)

Flood Treatment Variables (FEMA Definition)

Source: FEMA Disaster Declarations

Variable	Description
fe1	FEMA flood emergency declaration indicator
fe1_fl1h	FEMA severe flood indicator (flood + at least 1 housing damage)
fe1_fl1h_1st_year	Year of first severe FEMA flood (2000-2017), missing if never flooded
fe1_fl1h_2nd_year	Year of second severe FEMA flood, missing if < 2 floods
all_fe1_fl1h_1st_year	Year of first severe FEMA flood (including pre-2000)
never_fe1_fl1h	=1 if county never experienced severe FEMA flood

Flood Treatment Variables (NOAA Definition)

Source: NOAA Storm Events Database

Variable	Description
fl1d	NOAA flood with at least 1 direct death indicator
fl1h	NOAA flood with at least 1 housing damage indicator
fl1d_1st_year	Year of first NOAA flood death (2000-2017)
fl1d_2nd_year	Year of second NOAA flood death
all_fl1d_1st_year	Year of first NOAA flood death (including pre-2000)
never_fl1d	=1 if county never experienced NOAA flood death

Event Study Indicators

Pre-treatment leads (F) and post-treatment lags (L) for event study design.

Variable Pattern	Description
F_FH_{k}	FEMA: =1 if county is k years before first flood (k = 2,...,26)
L_FH_{k}	FEMA: =1 if county is k years after first flood (k = 0,...,26)
F_D_{k}	NOAA: =1 if county is k years before first NOAA flood death
L_D_{k}	NOAA: =1 if county is k years after first NOAA flood death

Note: In analysis, indicators beyond +/-10 are binned into endpoint bins (F_FH_10+ and L_FH_10+). The reference period is t = -1 (omitted).

Other FEMA Controls

Variable	Description
fema_allother	=1 if county received any non-flood FEMA declaration in current year
fema_allother_last1	=1 if county received non-flood FEMA declaration 1 year ago
fema_allother_last2	=1 if county received non-flood FEMA declaration 2 years ago
fema_allother_last3	=1 if county received non-flood FEMA declaration 3 years ago

County Classification Variables

Source: Appalachian Regional Commission

Variable	Description
ruralregion	=1 if rural Appalachian county (ARC classification)
ruralurbancode	USDA rural-urban continuum code
appasubregion	ARC sub-region designation
econcate2002	ARC economic status 2002 (Distressed/At-Risk/Transitional/Competitive/Attainment)
econcate2003	ARC economic status 2003
econcate2004	ARC economic status 2004
pctnation2000_medianhhi	County median HHI as percentile of national distribution (2000)
pctnation2000_unemploy	County unemployment as percentile of national distribution (2000)
pctnation2000_poverty	County poverty as percentile of national distribution (2000)

Generated Variables (created in analysis scripts)

These variables are NOT in the dataset but are created during analysis:

Variable	Created In	Description
death_{drug}_{pop}_rate	01	Mortality rate per 100,000
death_{drug}_{pop}_t	01	IHS transform of death count
death_{drug}_{pop}_rate_t	01	IHS transform of death rate
distressed	01	=1 if Distressed in 2002, 2003, or 2004
lowmdi	01	=1 if 2000 median HHI percentile $\leq$ 30.251
F_FH_1	01	Reference period indicator (set to 0)
stateencode	01	Numeric encoding of state
stateyearencode	01	Numeric encoding of state x year